

Module 18 Operations Console

IBM InfoSphere Information Server for Data Integration



Module Objectives

After completing this topic, you should be able to:

- Explain what is operations console
- Describe the operations console configuration and maintenance

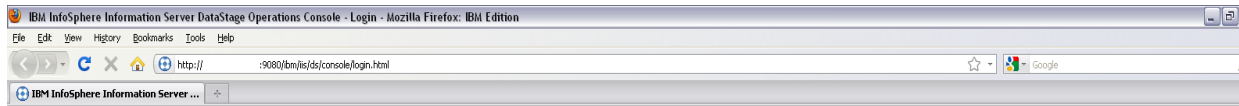
Operations Console - Introduction

- Provides a view into the operational environment of Information Server
- Simple, web based tool
- Server wide views of current job run activity
- High level summary and detailed information for both current and historical job runs
- Intuitive graphs illustrating job performance over time
- Visual cues alert user to potential issues
- Fault finding capabilities:
 - Analyze key Job run metrics
 - Compare properties of different Job runs (performance metrics, logs etc.)
 - Analyze operating system loading
 - Investigate system activity by “timeframe”
- Access control with read-only mode for operators and project level control
- Resource consumption views
- Backed by an open database containing all the operational data (published schema)

Operations Console – Platform Support

- Operations Console server components supported on all supported IS 8.7/9.1/9.1.2 platforms
- Operations database schema (ODB) supported in the following database environments:
 - IBM DB2 (9.5, 9.7, 10.1)
 - Oracle (10gR2m, 11gR2)
 - Microsoft SQLServer on a Windows (2005 sp2, 2008)
- Operations Console supports the following browsers:
 - Firefox 3.5 & 3.6
 - Internet Explorer 7 & 8
 - <http://<domain>:9080/ibm/iis/ds/console/login.html>
 - ...or...
 - <https://<domain>:9443/ibm/iis/ds/console/login.html>
 - Note that for other browsers and earlier versions of Firefox and Internet Explorer the browser check can be skipped by modifying the URL to include the parameter ignoreBrowserCheck, i.e.
<login.html?ignoreBrowserCheck=true>

Operations Console - Login



Login

Please enter your information

User name:

Password:

Please note that after some time of inactivity, the system will log you out automatically and ask you to sign in again.

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- The web application is accessed using a URL of the form:
- `http://<domain>:9080/ibm/iis/ds/console`
- Familiar IS login screen
- Login with IS credentials requires any DS role

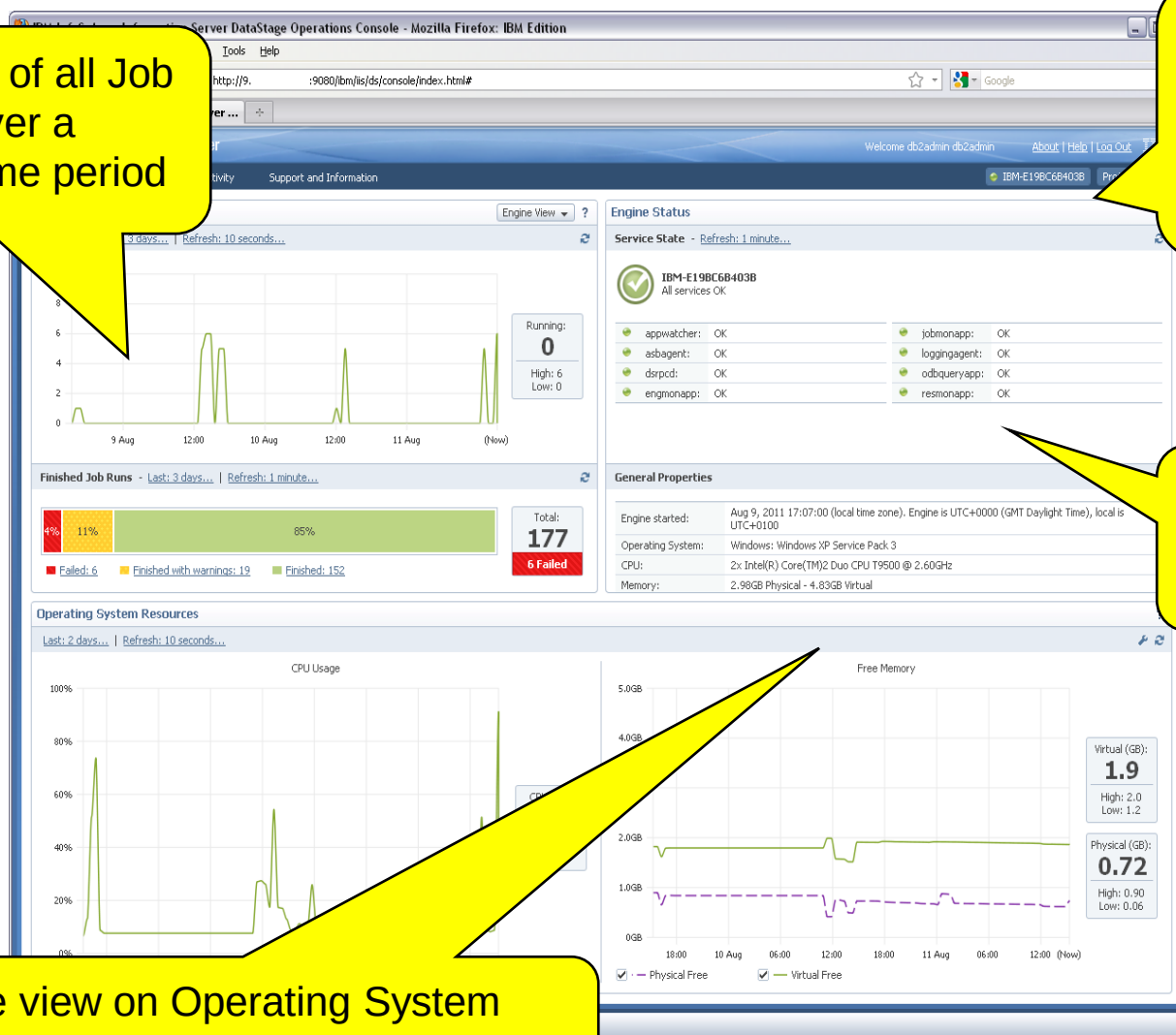
Operations Console – Home Page

High-level view of all Job Activity over a configurable time period

Currently selected Engine within the domain

Status of Engine components

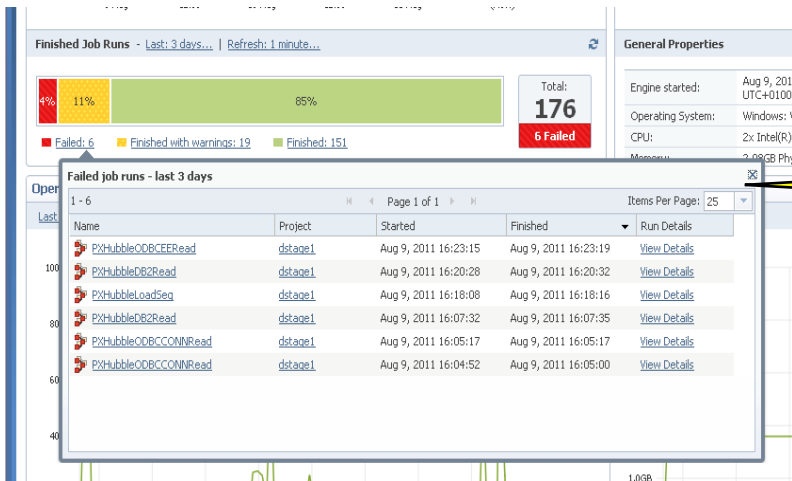
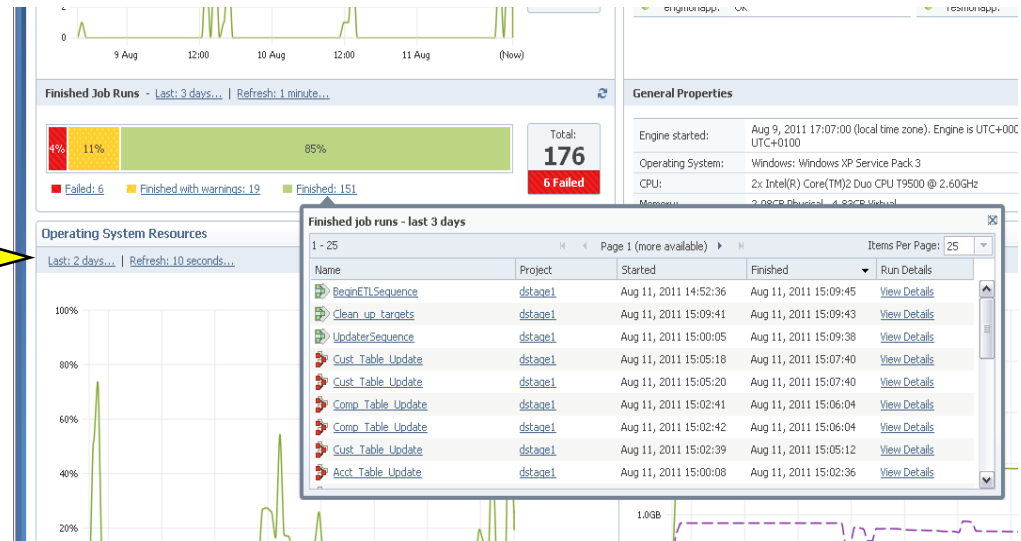
Configurable view on Operating System resources – can switch between conductor and any remote nodes if MPP



Operations Console – Job Run Activity

View recent Job run activity, via shortcuts:

- Finished
- Finished with warnings
- Failed



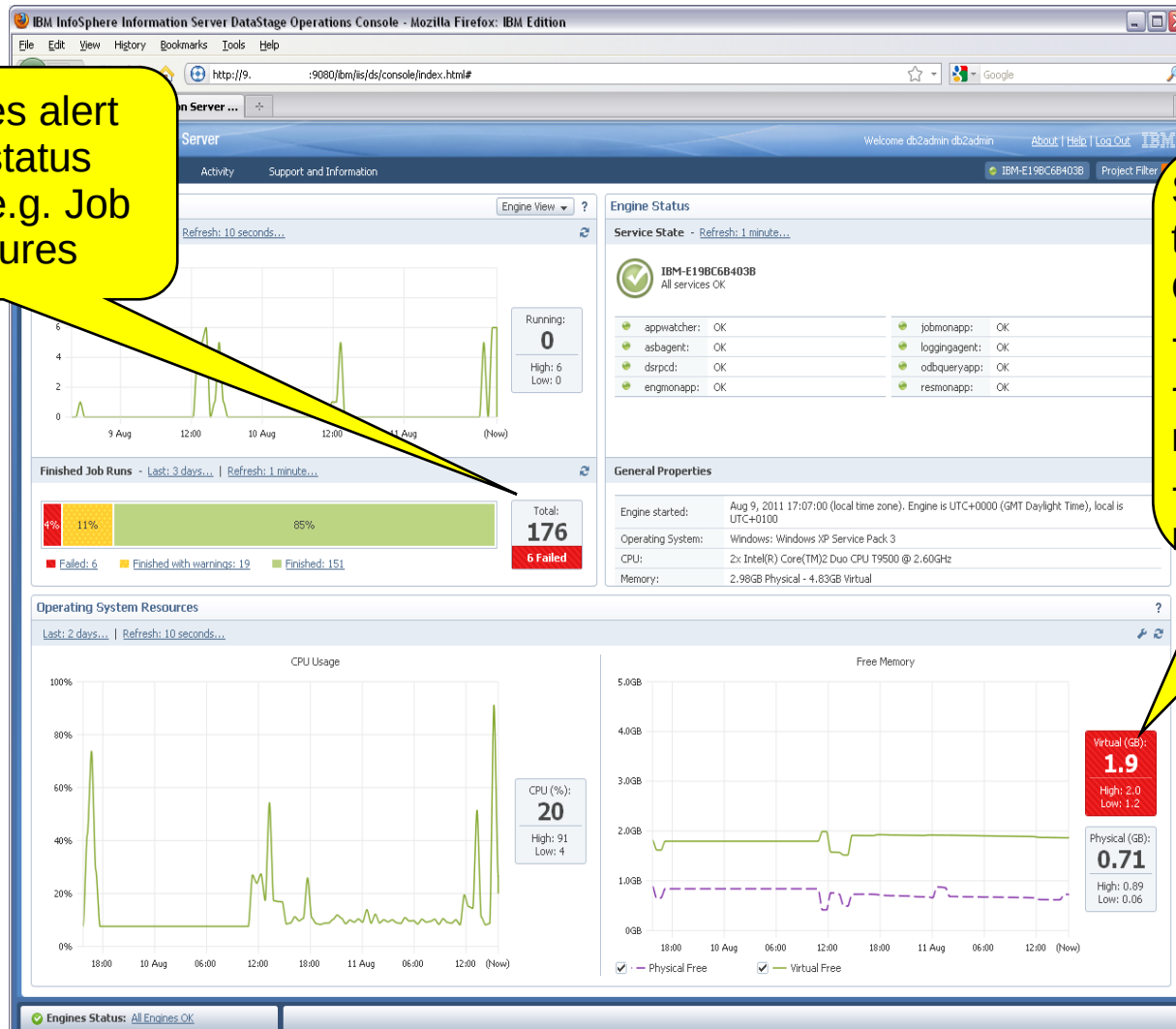
Select "View Details" to see details of the Job run

Operations Console - Alerts

Visual cues alert user to status concerns e.g. Job Run failures

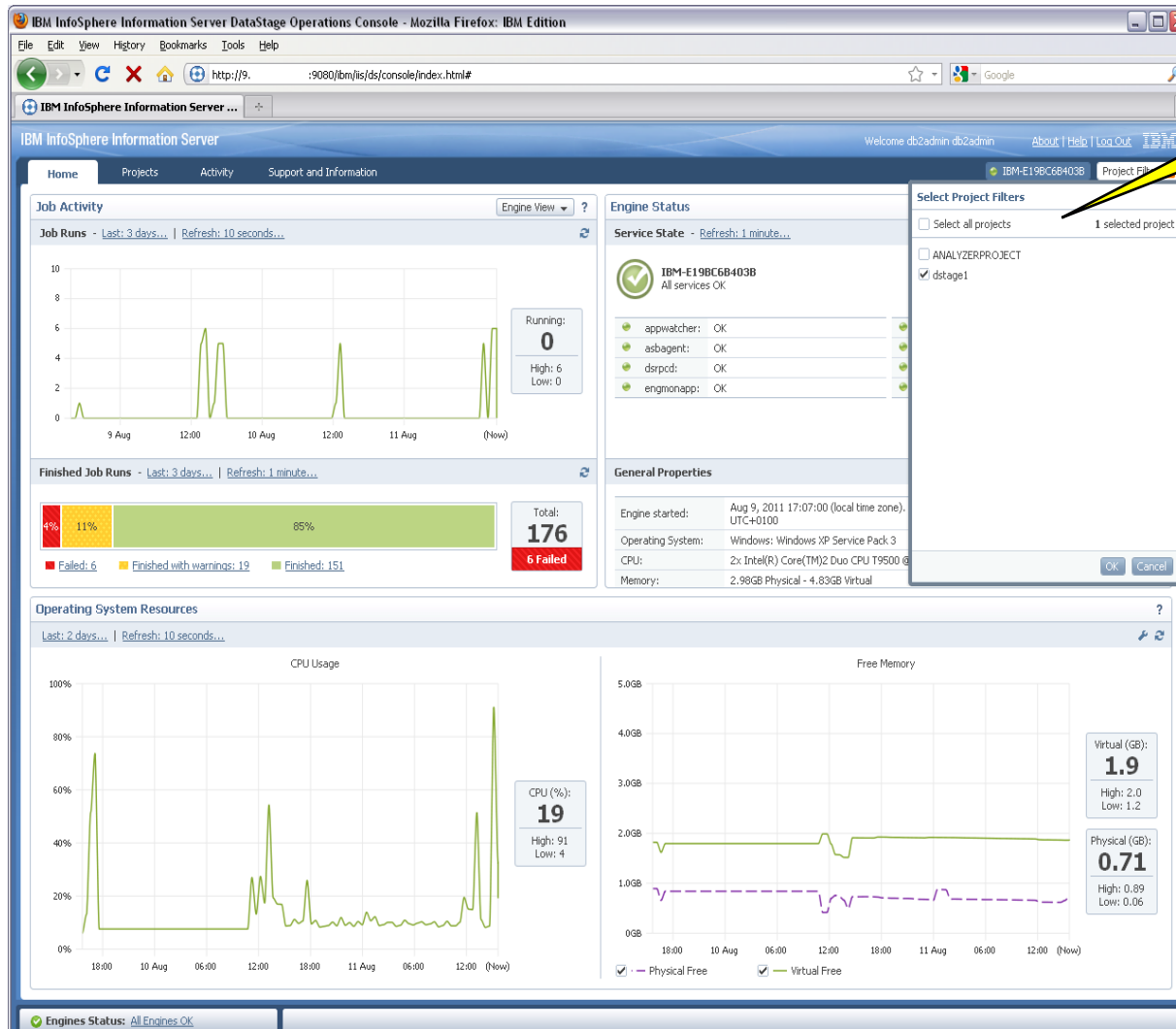
Set Alert thresholds for key OS metrics:

- CPU
- Free virtual memory
- Free physical memory



Operations Console – Project Filter

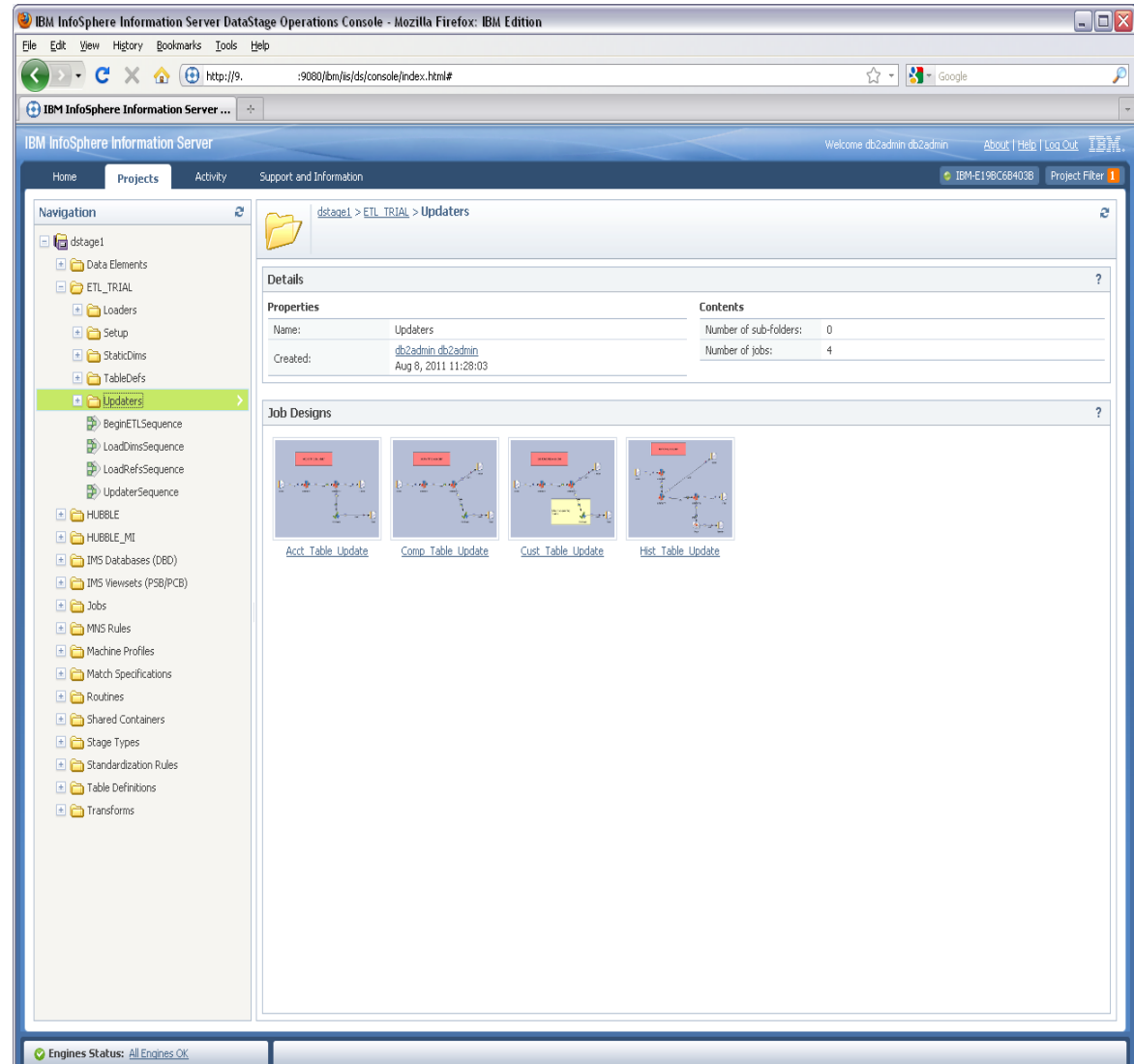
Filter display
by Engine or
Project(s)



Note: Access control prevents display of data from projects which the current user does not have access to, as determined by their entitled projects.

Operations Console – Projects Page

- Browse Project content
 - Folder content
 - Thumbnails of job designs
 - Project properties
 - Environment variables



Operations Console – Projects Page – Job details

Job properties

Runs of this job

The screenshot displays the IBM InfoSphere Information Server DataStage Operations Console in a Mozilla Firefox browser. The interface includes a navigation pane on the left, a main content area with tabs for 'Home', 'Projects', 'Activity', and 'Support and Information', and a status bar at the bottom.

Navigation Pane: Shows a tree structure of the project 'dstage1'. The 'BeginETLSequence' job is highlighted under the 'Loaders' folder.

Job Details: The 'Details' tab is active, showing the following properties:

Property	Value
Name	BeginETLSequence
Short description	Main Sequence
Created	db2admin_db2admin Aug 8, 2011 11:28:04
Last modified	db2admin_db2admin Aug 8, 2011 11:32:25
Last compiled	Aug 8, 2011 10:32:26
Job type	Sequence
Multi-instance	No
Information Services enabled	No
Used by job(s)	0
Depends on job(s)	S

Latest Activity:

Property	Value
Last started	db2admin_db2admin Aug 11, 2011 14:52:36
Invocation ID	
Status	Finished, Aug 11, 2011 15:09:45
Elapsed	00:17:09
CPU time	00:00:07
Rows in	(Not applicable)
Rows out	(Not applicable)
Messages	25

Job Runs: The 'Job Runs' tab is active, showing a list of job runs. The table below contains the data for the first 8 runs.

Started	Finished	Invocation ID	Status	Elapsed	Rows In / Second	Rows Out / Second
Aug 11, 2011 14:52:36	Aug 11, 2011 15:09:45		Finished	00:17:09		
Aug 11, 2011 13:01:02	Aug 11, 2011 13:02:53		Finished	00:01:51		
Aug 10, 2011 13:01:11	Aug 10, 2011 13:03:44		Finished	00:02:33		
Aug 9, 2011 16:38:40	Aug 9, 2011 16:40:49		Finished	00:02:09		
Aug 9, 2011 16:12:15	Aug 9, 2011 16:14:01		Finished	00:01:46		
Aug 9, 2011 13:01:05	Aug 9, 2011 13:02:59		Finished	00:01:54		
Aug 8, 2011 14:01:02	Aug 8, 2011 14:02:58		Finished	00:01:56		
Aug 8, 2011 11:33:41	Aug 8, 2011 11:35:47		Finished	00:02:06		

Engines Status: All Engines OK

Operations Console – Activity page

- Analyze Job run activity, either current or historical
 - List of current/recent Job runs
 - Performance metrics (rows processed, elapsed time)
 - Status information

IBM InfoSphere Information Server DataStage Operations Console - Mozilla Firefox: IBM Edition

http://9. :9080/ibm/is/console/index.html#

Welcome db2admin db2admin About | Help | Log Out

IBM InfoSphere Information Server

Home Projects Activity Support and Information

Current Activity Past Activity View: 5 Days

Job Runs Resources

View Details | Refresh: 10 seconds...

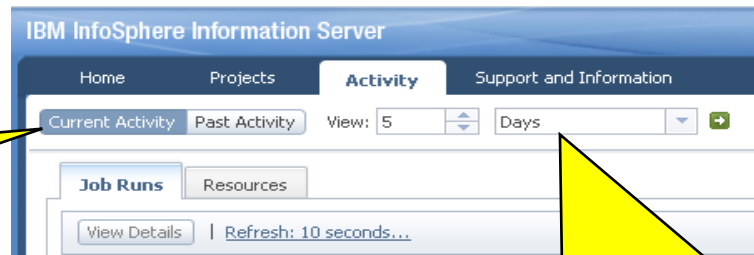
1 - 25 Page 1 (more available)

Name	Project	Invocation ID	Started	Finished	Status	Elapsed	Rows In / Second	Rows Out / Second
Information Services (0)								
Running (0)								
Completed (25)								
Cust_Table_Update	dstage1	CUST2	Aug 10, 2011 13:03:21	Aug 10, 2011 13:03:30	Finished	00:00:09	21	21
Comp_Dtm_load	dstage1		Aug 11, 2011 14:53:32	Aug 11, 2011 14:59:10	Finished	00:05:38	12967	12881
LoadRefsSequence	dstage1		Aug 11, 2011 13:01:53	Aug 11, 2011 13:02:13	Finished	00:00:20		
Comp_Table_Update	dstage1	COMP2	Aug 11, 2011 15:02:41	Aug 11, 2011 15:06:04	Finished	00:03:23	21233	21233
P2HubbleCreateSeqMI	dstage1	INV6	Aug 9, 2011 13:59:08	Aug 9, 2011 13:59:15	Finished	00:00:07	143	143
Acct_Table_Update	dstage1	ACCT1	Aug 9, 2011 16:39:58	Aug 9, 2011 16:40:05	Finished	00:00:07	22335	22335
Comp_Table_Update	dstage1	COMP1	Aug 10, 2011 13:03:17	Aug 10, 2011 13:03:29	Finished	00:00:12	0	0
P2HubbleCreateSeq	dstage1		Aug 8, 2011 16:29:40	Aug 8, 2011 16:29:56	Finished	00:00:16	63	63
P2HubbleCreateSeqMI	dstage1	INV1	Aug 9, 2011 14:00:36	Aug 9, 2011 14:00:44	Finished	00:00:08	125	125
Create_Time_Dtm	dstage1		Aug 8, 2011 11:35:00	Aug 8, 2011 11:35:06	Finished	00:00:06	14400	14400
LoadRefsSequence	dstage1		Aug 9, 2011 16:13:00	Aug 9, 2011 16:13:20	Finished	00:00:20		
Acct_Table_Update	dstage1	ACCT2	Aug 8, 2011 11:35:11	Aug 8, 2011 11:35:18	Finished	00:00:07	49	49
LoadDmsSequence	dstage1		Aug 9, 2011 13:01:19	Aug 9, 2011 13:01:56	Finished	00:00:37		
Cust_Table_Update	dstage1	CUST2	Aug 9, 2011 16:13:41	Aug 9, 2011 16:13:49	Finished	00:00:08	24	24
LoadRefsSequence	dstage1		Aug 9, 2011 16:39:36	Aug 9, 2011 16:39:55	Finished	00:00:19		
BeginMTLSequence	dstage1		Aug 9, 2011 16:38:40	Aug 9, 2011 16:40:49	Finished	00:02:09		
Create_Time_Dtm	dstage1		Aug 9, 2011 16:13:13	Aug 9, 2011 16:13:19	Finished	00:00:06	14400	14400
Comp_Table_Update	dstage1	COMP1	Aug 8, 2011 14:02:34	Aug 8, 2011 14:02:42	Finished	00:00:08	60	60
P2HubbleCreateSeqMI	dstage1	INV8	Aug 9, 2011 14:00:46	Aug 9, 2011 14:00:54	Finished	00:00:08	125	125
P2HubbleCreateSeqMI	dstage1	INV1	Aug 9, 2011 13:58:54	Aug 9, 2011 13:59:07	Finished	00:00:13	77	77
Acct_Table_Update	dstage1	ACCT1	Aug 11, 2011 13:02:16	Aug 11, 2011 13:02:22	Finished	00:00:06	57	57
Cust_Table_Update	dstage1	COMP3	Aug 11, 2011 15:02:39	Aug 11, 2011 15:05:12	Finished	00:02:33	9786	9786
Cust_Table_Update	dstage1	CUST2	Aug 11, 2011 13:02:34	Aug 11, 2011 13:02:41	Finished	00:00:07	27	27
P2HubbleCreateSeqMI	dstage1	INV2	Aug 9, 2011 14:04:38	Aug 9, 2011 14:04:44	Finished	00:00:06	167	167
P2HubbleCreateSeq	dstage1		Aug 8, 2011 15:34:40	Aug 8, 2011 15:34:47	Finished	00:00:07	143	143

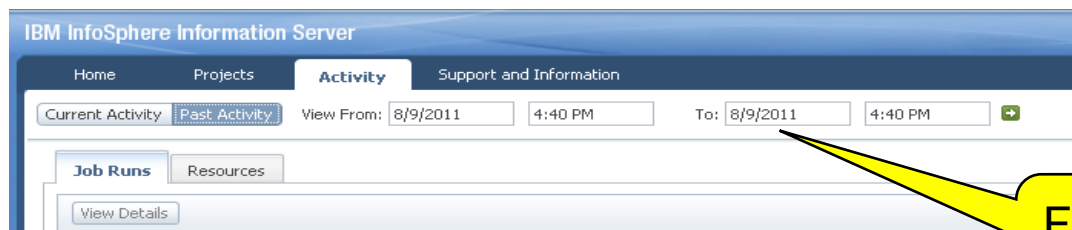
Engines Status: All Engines OK

Operations Console – Activity Page – Time period

Switch between
“current” or “past”
activity



For “current”, select the time
period to include (time from now)

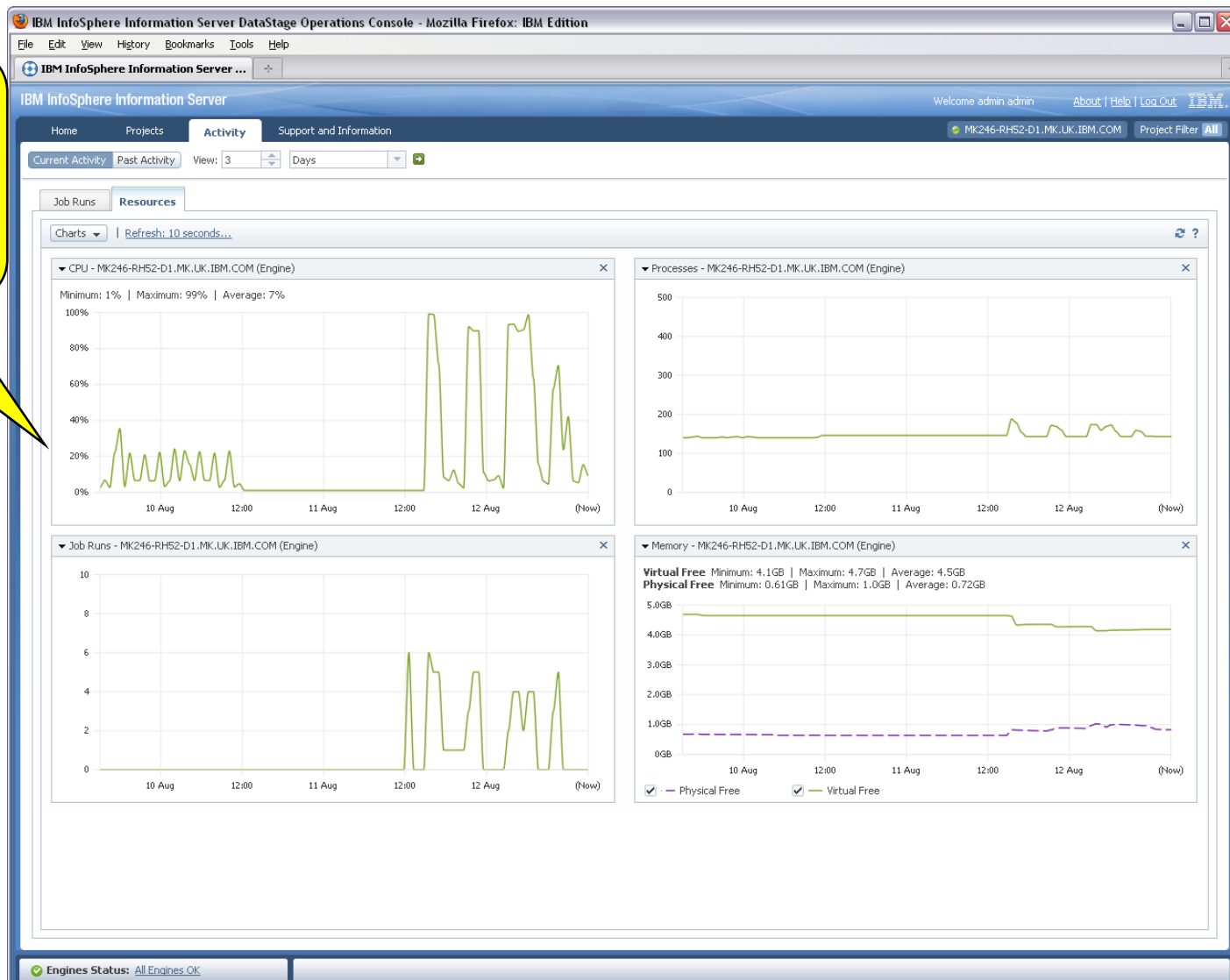


For “past”, define the timeframe
to include

Operations Console – Activity Page – Resources

Can switch between conductor and any remote nodes if MPP/grid

- More detailed metrics are captured in the ODB than shown in the UI at present



Operations Console – Job Run Analysis

- Drill down into the Job run information for analysis
- Analyze job runs individually or compare different runs of the same job
- Job run information includes:
 - Properties of the Job
 - Job parameters used for that run
 - Job runtime metrics (elapsed time, CPU used, rows processed)
 - System metrics at the time the Job ran (system CPU, memory, disk space)
 - Log messages (subset including control, warning and fatal messages)
 - Configuration file details for parallel jobs
- Job runs can also be analyzed in the context of what other Jobs are running on the system at that time or within a related time period

Operations Console – Job Analysis - Run Details

IBM InfoSphere Information Server DataStage Operations Console - Mozilla Firefox: IBM Edition

File Edit View History Bookmarks Tools Help

http://9. :9080/ibm/iis/ds/console/index.html#

IBM InfoSphere Information Server

Welcome db2admin db2admin About Help

Home Projects Activity Support and Information IBM-E19B...

Job Activity

Job Runs - Last: 9 Aug

Finished Job Runs

Operating System

Last: 2 days...

100%

Run Details: PXHubbleODBCERead

Details Performance Log Messages

Properties

Name: [PXHubbleODBCERead](#)

Project: [dstage1](#)

Executable compiled on: Aug 9, 2011 16:23:11

Job type: Parallel

Run type: Normal

Started by job run:

Started job run(s): [View List](#)

Configuration file: [C:\IBM\InformationServer\Server\Configurations\default.apf](#)

Conductor PID: 4540

Master PID: 6072

Started: [db2admin db2admin](#)
Aug 9, 2011 16:23:15

Invocation ID:

Status: Failed, Aug 9, 2011 16:23:19

Elapsed: 00:00:04

CPU time: 00:00:00

Rows in: 0 (0 per second)

Rows out: 0 (0 per second)

Messages: 14 (Fatal: 3, warnings: 1)

Parameters

Name Value

XMETAUSER xmeta

XMETAPASS **

\$APT_CONFIG_FILE C:\

Run Details: PXHubbleODBCERead

Details Performance Log Messages

1 - 5 Page 1 of 1 Items Per Page: 25

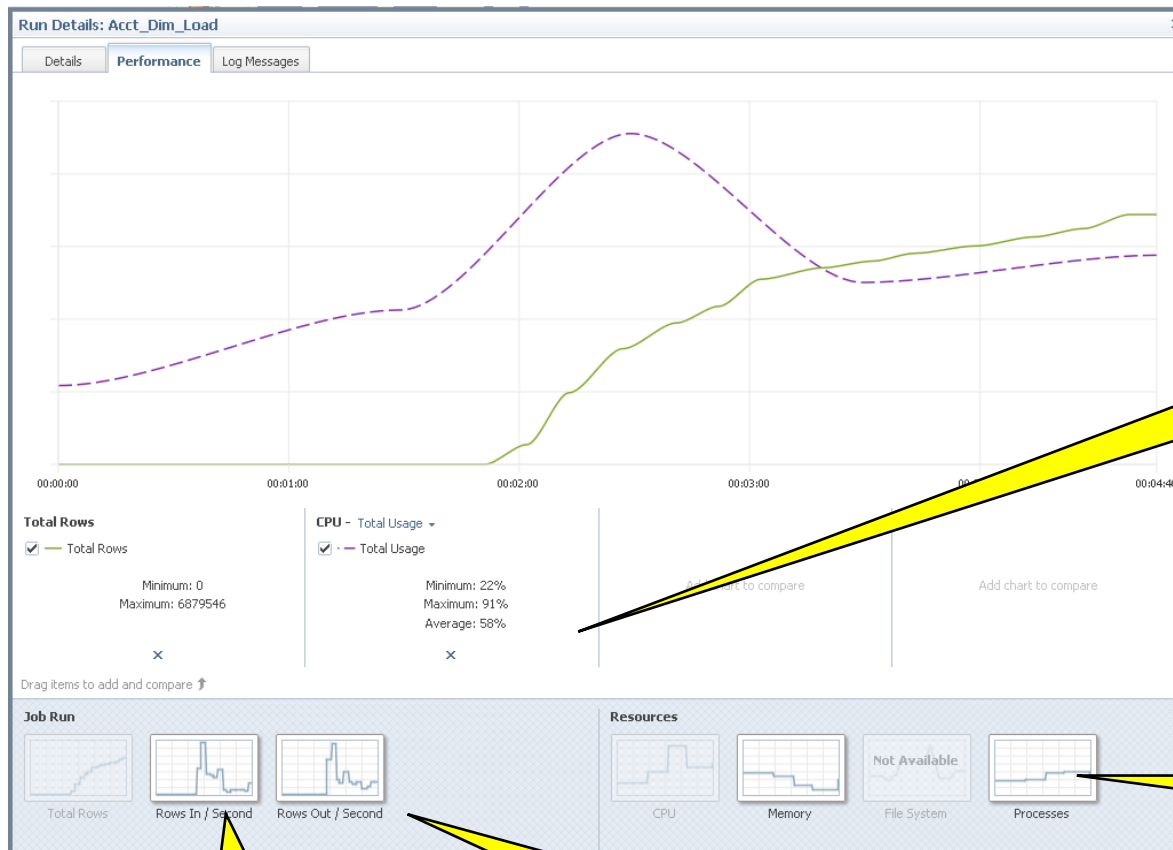
Elapsed	Message
00:00:01	Starting Job PXHubbleODBCERead.
00:00:01	Environment variable settings: (...)
00:00:03	main_program: [IBM][CLI Driver][DB2/NT] SQL0204N "DB2XMETA.DATASTAGEX_XMETAGEN_DSFLOWVARIABLEC2E76D84" is an undefined name. SQLSTATE=42704
00:00:04	ODBC_EE: When checking operator: The modify operator has a binding for the non-existent output field "XMETA_REPO5_OBJECT_ID_XMETA".
00:00:04	Job PXHubbleODBCERead aborted.

Details shows Job properties and information about the run

Click links for more info

View Job log messages

Operations Console – Job Run Performance



Can switch between conductor and compute nodes if configured.

Available metrics – drag to add to main area

Graph profile of actual data

Rows in/out/total are summed for all stages that are sources or targets for the job, including lookups – i.e. external connections

Operations Console – Run Comparison

Job Runs

[View Details](#) [Investigate Timeframe](#)

1 - 24

Page 1 of 1

Started	Finished	Invocation ID	Status	Elapsed	Rows In / Second	Rows Out / Second
Aug 11, 2011 15:05:18	Aug 11, 2011 15:07:40	CUST2	✓ Finished	00:02:22	10544	10544
Aug 11, 2011 15:05:20	Aug 11, 2011 15:07:40	CUST1	✓ Finished	00:02:20	10695	10695
Aug 11, 2011 15:02:39	Aug 11, 2011 15:05:12	COMP3	✓ Finished	00:02:33	9786	9786
Aug 11, 2011 13:02:36	Aug 11, 2011 13:02:43	CUST1	✓ Finished	00:00:07	27	27
Aug 11, 2011 13:02:34	Aug 11, 2011 13:02:41	CUST2	✓ Finished	00:00:07	27	27
Aug 11, 2011 13:02:25	Aug 11, 2011 13:02:32	COMP3	✓ Finished	00:00:07	27	27

Comparison of Run Details: Cust_Table_Update

[Details](#)
[Performance](#)
[Log Messages](#)

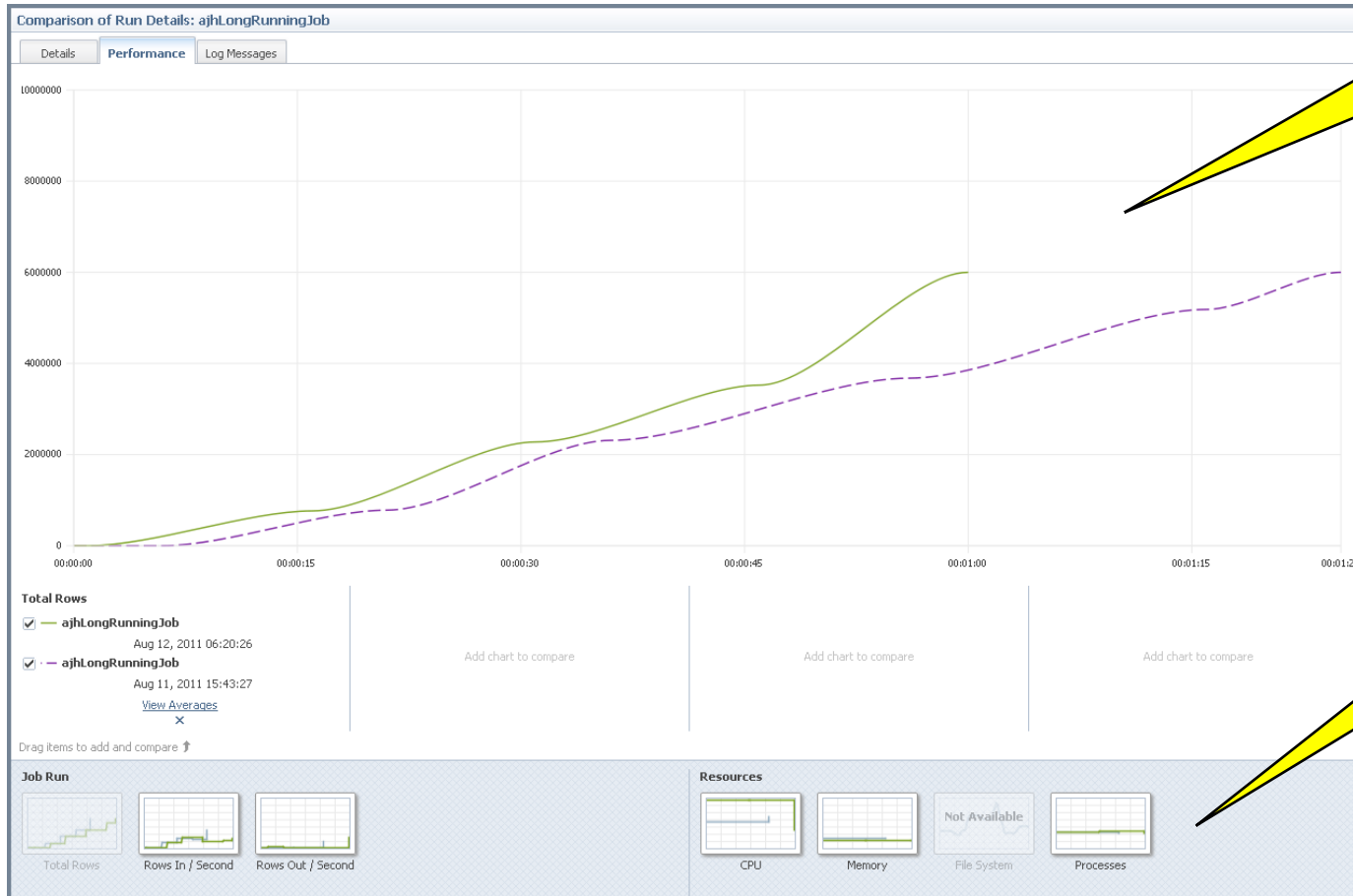
Properties
☒ Only display comparison differences (6)

Cust_Table_Update.CUST1 - Aug 11, 2011 15:05:20 Started job run(s): View List Elapsed: 00:02:20 CPU time: 00:00:16 Rows in: 1497254 (10695 per second) Rows out: 1497254 (10695 per second) Messages: 27	Cust_Table_Update.CUST1 - Aug 11, 2011 13:02:36 View List 00:00:07 00:00:00 190 (27 per second) 190 (27 per second) 17
---	---

Parameters
☒ Only display comparison differences (1)

Cust_Table_Update.CUST1 - Aug 11, 2011 15:05:20 Delay: 100	Cust_Table_Update.CUST1 - Aug 11, 2011 13:02:36 0
--	---

Operations Console – Performance comparison



Rows
processed for
each run

Available
metrics

Operations Console – Log Comparison

Log for successful run data

Log for failed run

Comparison of Run Details: PXHubbleDB2Read

Details Performance **Log Messages**

PXHubbleDB2Read - Aug 9, 2011 16:21:26

1 - 3 Page 1 of 1 Items Per Page: 25

Elapsed	Message
00:00:01	Starting Job PXHubbleDB2Read.
00:00:01	Environment variable settings: (...)
00:00:06	Finished Job PXHubbleDB2Read.

PXHubbleDB2Read - Aug 9, 2011 16:20:28

1 - 9 Page 1 of 1 Items Per Page: 25

Elapsed	Message
00:00:00	Starting Job PXHubbleDB2Read.
00:00:01	Environment variable settings: (...)
00:00:04	DB2_UDB_MASTERREF: Error Idx = 5; (...)
00:00:04	DB2_UDB_MASTERREF: Prepare failed
00:00:04	DB2_UDB_MASTERREF: When checking operator: The modify operator has a binding for the non-existent out
00:00:04	DB2_UDB_MASTERREF: When checking operator: The modify operator has a binding for the non-existent out
00:00:04	DB2_UDB_MASTERREF: When checking operator: The modify operator has a binding for the non-existent out
00:00:04	DB2_UDB_MASTERREF: When checking operator: The modify operator has a binding for the non-existent out
00:00:04	Job PXHubbleDB2Read aborted.

Aug 9, 2011 16:20:32
Message ID: [IIS-DSEE-TNDB-00534](#)
DB2_UDB_MASTERREF: Error Idx = 5;
DB2Driver Embedded SQL message: SQL0204N "DSODB.MASTERREF" is an undefined name. SQLSTATE=42704
;
sqlcode = -204;
sqlstate = 42704

Log message detail for
Fatal error

Message ID hyperlinked to
help system

Operations Console – Basic Configuration Steps

- Operations Console components are installed with DataStage, but not pre-configured to run.
- The operations database needs to be created manually before configuring the Operations Console components. Scripts are provided to create the database.
- To enable monitoring on an engine system, you must do the following steps after installing the engine tier:
 - Set up a schema and tables in the operations database (ODB), or identify an existing operations database which this system can share.
 - Configure the engine system monitor to connect to its chosen ODB.
 - Configure the monitor so it knows what data to collect, and how frequently, if you do not want to use the defaults.
 - Ensure that the monitor server processes are started on each engine and that monitoring is switched on.
- If these steps have not been performed, the engine tier is available for use, but monitoring is not enabled.
- There are various readme files available in Server/DSODB and Server/DSODB/scripts/* as well as documentation on the 8.7/9.1/9.1.2 InfoCenter - “Getting started with the Operations Console”

Operations Console – Configure Connection

- The Configuration file DSODBCConnect.cfg must be edited to specify:
 - Which DBMS type
 - The file located at Server/DSODB/scripts/<dbms+version>
 - JAR to use for JDBC
 - By defaults, JARs for DB2, Oracle and SQLServer are included
 - URL to identify database
 - If not xmeta!
 - User name and password
 - Can be encrypted.

Ops DB Maintenance – Purging & Reporting

- The operations database (ODB) will require maintenance in respect of its size.
- The flexibility provided by the ODB and various tuning parameters mean that large volumes of operational data can be captured and stored over time.
- A command-line tool is provided via the istool ODBAdmin command.
- ODBAdmin supports two main features, “countdb” and “purgedb”
- “countdb” returns the size of the data based on the specified parameters (e.g. by date)
- “purgedb” deletes the data based on the specified parameters

Ops DB Maintenance – Backup & Restore

- If you co-locate the operations database with the metadata repository it could be covered by the same backup schedule. A separately installed database would need its own backup schedule.
- Monitoring services should be stopped in the console while backup is in progress
- Jobs can still run but progress will not be visible in console UI
- Job events will catch up once monitoring services started again
- System resource snapshots will be lost for the period when monitoring services were stopped.